

Estimation of the Number of Vacancies

1 Notation and Data

We consider a finite population of job offers $U = \{1, 2, \dots, N\}$, where N is unknown, and three job portals. For $i \in U$, we define:

- $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, Y_{i3}) \in \{0, 1\}^3$ – the capture history,
- $G_i \in \{1, 2, \dots, 9\}$ – the true (often unknown) occupation code,
- $X_i \in \{1, 2\}$ – the latent class,
- $V_i \in \{1, 2, \dots\}$ – the number of vacancies in a job offer.

We want to estimate the total number of vacancies, i.e.,

$$T = \sum_{i=1}^N V_i. \quad (1)$$

Units with $\mathbf{Y}_i = (0, 0, 0)$ are unobserved. Let S denote the set of observed job offers, i.e.,

$$S := \{i \in U : \mathbf{Y}_i \neq (0, 0, 0)\}. \quad (2)$$

Let

$$N_{\mathbf{y},g,x} := \#\{i \in U : \mathbf{Y}_i = \mathbf{y}, G_i = g, X_i = x\}. \quad (3)$$

Furthermore, let $\tilde{G}_i$ denote the observed occupation code. If $Y_{i1} = 1$, then $\tilde{G}_i = G_i$. Otherwise, the observed code might differ from the true code.

Let $\tilde{V}_i$ denote the observed number of vacancies in offer i . If $Y_{i1} = 1$, then $\tilde{V}_i = V_i$. Otherwise, information about V_i might be missing or censored at 2+.

Additionally, we assume the existence of an independent validation sample, denoted by S_{val} , which is disjoint from the main set of job offers S . For the units in S_{val} , both the true occupation code G_i and the observed (predicted) code $\tilde{G}_i$ are known.

2 Assumptions

(A1) The population is closed.

(A2) There are no linkage errors.

(A3) There exists a constant confusion matrix $\mathbf{\Pi} \in [0, 1]^{9 \times 9}$ such that for the units $i \in S$, we have

$$\mathbb{P}(\tilde{G}_i = h \mid G_i = g, Y_{i1} = 0) = \Pi_{gh}, \quad g, h \in \{1, 2, \dots, 9\}. \quad (4)$$

(A4) For each combination $(\mathbf{y}, g, x)$, there exists an expected cell frequency $m_{\mathbf{y},g,x} > 0$, and the cell counts are independent and follow a Poisson distribution, i.e.,

$$N_{\mathbf{y},g,x} \sim \text{Poisson}(m_{\mathbf{y},g,x}). \quad (5)$$

We consider the following log-linear model (cf. Baffour et al. (2021)):

$$\log m_{\mathbf{y},g,x} = \lambda + \lambda_g^G + \lambda_x^X + \sum_{s=1}^3 \lambda_{y_s}^{(s)} + \sum_{s=1}^3 \lambda_{y_s,x}^{(sX)} + \lambda_{g,x}^{(GX)} + \lambda_{y_1,y_2}^{(12)} + \lambda_{y_1,y_3}^{(13)}. \quad (6)$$

(A5) Given $G_i = g$, V_i follows a zero-truncated negative binomial distribution with parameters μ_g and σ (cf. Brännäs (1992)).

(A6) Given $G_i = g$, the observation/coarsening mechanism generating $\tilde{V}_i$ is ignorable (missing/coarsened at random).

(A7) Given $G_i = g$, V_i is independent of $\mathbf{Y}_i$ and X_i .

(A8) Given $G_i = g$ and $\mathbf{Y}_i$, $\tilde{G}_i$ and $\tilde{V}_i$ are independent of each other and of X_i .

3 Estimation of the Confusion Matrix

Using the validation sample S_{val} , we compute $\hat{\mathbf{\Pi}}$ as an estimate of the confusion matrix $\mathbf{\Pi}$. Its elements are given by:

$$\hat{\Pi}_{g,\tilde{g}} = \frac{\#\{i \in S_{val} : G_i = g, \tilde{G}_i = \tilde{g}\}}{\#\{i \in S_{val} : G_i = g\}}. \quad (7)$$

4 Likelihood Function

Let $f_g(\cdot; \mu_g, \sigma)$ denote the probability mass function of a zero-truncated negative binomial distribution with parameters μ_g and σ . For $i \in S$ and code g , we define

$$L_V(i \mid g; \mu_g, \sigma) = \begin{cases} f_g(\tilde{V}_i; \mu_g, \sigma), & \text{if } \tilde{V}_i = V_i, \\ \mathbb{P}_{\mu_g, \sigma}(V_i \geq 2 \mid G_i = g), & \text{if } \tilde{V}_i = 2+, \\ 1, & \text{if } \tilde{V}_i \text{ is missing.} \end{cases} \quad (8)$$

Let $\boldsymbol{\theta}$ denote the parameters associated with model (6), and let $\boldsymbol{\phi}$ denote the parameters associated with modeling the number of vacancies.

In our formulation, the capture part is treated through a complete-data log-likelihood based on the latent cell counts $N_{\mathbf{y},g,x}$, whereas the vacancy part is written directly in terms of the observed/coarsened vacancy data through $L_V(i \mid g; \boldsymbol{\phi})$. Therefore, conditional on $\hat{\boldsymbol{\Pi}}$, the objective function used in the EM algorithm consists of two components, given by:

$$\ell_c^{(Y)}(\boldsymbol{\theta}) = \sum_{\mathbf{y},g,x} [N_{\mathbf{y},g,x} \log m_{\mathbf{y},g,x}(\boldsymbol{\theta}) - m_{\mathbf{y},g,x}(\boldsymbol{\theta})] + \text{const} \quad (9)$$

and

$$\ell^{(V)}(\boldsymbol{\phi}) = \sum_{i \in S} \log L_V(i \mid G_i; \boldsymbol{\phi}). \quad (10)$$

5 EM Algorithm

5.1 E-step

In the t -th iteration, we want to calculate

$$Q(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}) = \mathbb{E}[\ell_c^{(Y)}(\boldsymbol{\theta}) + \ell^{(V)}(\boldsymbol{\phi}) \mid \{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}]. \quad (11)$$

Let us define

$$\tau_{i,g,x}^{(t)} := \mathbb{P}(G_i = g, X_i = x \mid \mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}). \quad (12)$$

Lemma 1. *The weights $\tau_{i,g,x}^{(t)}$ are given by*

$$\tau_{i,g,x}^{(t)} = \frac{\mathbb{P}(\tilde{G}_i \mid G_i = g, \mathbf{Y}_i; \hat{\boldsymbol{\Pi}}) \cdot L_V(i \mid g; \boldsymbol{\phi}^{(t)}) \cdot m_{\mathbf{Y}_i,g,x}(\boldsymbol{\theta}^{(t)})}{\sum_{g'=1}^9 \sum_{x'=1}^2 \mathbb{P}(\tilde{G}_i \mid G_i = g', \mathbf{Y}_i; \hat{\boldsymbol{\Pi}}) \cdot L_V(i \mid g'; \boldsymbol{\phi}^{(t)}) \cdot m_{\mathbf{Y}_i,g',x'}(\boldsymbol{\theta}^{(t)})}. \quad (13)$$

Proof. From Bayes' theorem, we have

$$\tau_{i,g,x}^{(t)} = \frac{\overbrace{\mathbb{P}(\tilde{G}_i, \tilde{V}_i \mid G_i = g, X_i = x, \mathbf{Y}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}})}^{(*)} \cdot \overbrace{\mathbb{P}(G_i = g, X_i = x \mid \mathbf{Y}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}})}^{(\dagger)}}{\underbrace{\mathbb{P}(\tilde{G}_i, \tilde{V}_i \mid \mathbf{Y}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}})}_{(\ddagger)}}. \quad (14)$$

By Assumptions (A7) and (A8), we obtain

$$(*) = \mathbb{P}(\tilde{G}_i \mid G_i = g, \mathbf{Y}_i; \hat{\boldsymbol{\Pi}}) \cdot \mathbb{P}(\tilde{V}_i \mid G_i = g; \boldsymbol{\phi}^{(t)}) = \mathbb{P}(\tilde{G}_i \mid G_i = g, \mathbf{Y}_i; \hat{\boldsymbol{\Pi}}) \cdot L_V(i \mid g; \boldsymbol{\phi}^{(t)}). \quad (15)$$

Furthermore, under Assumption (A4), the probability that a specific unit with the observed capture history $\mathbf{Y}_i$ belongs to the latent class x and the true occupation code g is proportional to the expected frequency of that cell within the $\mathbf{Y}_i$ stratum, i.e.,

$$(\dagger) = \frac{m_{\mathbf{Y}_i,g,x}(\boldsymbol{\theta}^{(t)})}{\sum_{g'=1}^9 \sum_{x'=1}^2 m_{\mathbf{Y}_i,g',x'}(\boldsymbol{\theta}^{(t)})}. \quad (16)$$

Finally, by the law of total probability, we have

$$\begin{aligned}
(\ddagger) &= \sum_{g'=1}^9 \sum_{x'=1}^2 \left[\mathbb{P}(\tilde{G}_i, \tilde{V}_i \mid G_i = g', X_i = x', \mathbf{Y}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}) \cdot \mathbb{P}(G_i = g', X_i = x' \mid \mathbf{Y}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}) \right] \\
&= \sum_{g'=1}^9 \sum_{x'=1}^2 \left[\mathbb{P}(\tilde{G}_i \mid G_i = g', \mathbf{Y}_i; \hat{\Pi}) \cdot L_V(i \mid g'; \boldsymbol{\phi}^{(t)}) \cdot \frac{m_{\mathbf{Y}_i, g', x'}(\boldsymbol{\theta}^{(t)})}{\sum_{g''=1}^9 \sum_{x''=1}^2 m_{\mathbf{Y}_i, g'', x''}(\boldsymbol{\theta}^{(t)})} \right].
\end{aligned} \tag{17}$$

Substituting $(*)$, $(\dagger)$, and $(\ddagger)$ back into (14) yields the desired result. $\square$

Remark 1. By Assumption (A3), $\mathbb{P}(\tilde{G}_i \mid G_i = g, \mathbf{Y}_i; \hat{\Pi})$ equals $\hat{\Pi}_{g, \tilde{G}_i}$ if $Y_{i1} = 0$, and equals 1 if $Y_{i1} = 1$ and $\tilde{G}_i = g$ (and 0 otherwise).

To obtain (11), we calculate $N_{\mathbf{y}, g, x}^{(t)} = \mathbb{E}[N_{\mathbf{y}, g, x} \mid \{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}]$ as shown in Lemma 2.

Lemma 2. We have

$$N_{\mathbf{y}, g, x}^{(t)} = \begin{cases} \sum_{i \in S: \mathbf{Y}_i = \mathbf{y}} \tau_{i, g, x}^{(t)}, & \text{if } \mathbf{y} \neq (0, 0, 0), \\ m_{\mathbf{y}, g, x}(\boldsymbol{\theta}^{(t)}), & \text{if } \mathbf{y} = (0, 0, 0). \end{cases} \tag{18}$$

Proof. Assume that $\mathbf{y} \neq (0, 0, 0)$. We have

$$\begin{aligned}
N_{\mathbf{y}, g, x}^{(t)} &= \sum_{i \in S: \mathbf{Y}_i = \mathbf{y}} \mathbb{E}[\mathbb{I}(G_i = g, X_i = x) \mid \mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}] \\
&= \sum_{i \in S: \mathbf{Y}_i = \mathbf{y}} \mathbb{P}(G_i = g, X_i = x \mid \mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}) = \sum_{i \in S: \mathbf{Y}_i = \mathbf{y}} \tau_{i, g, x}^{(t)}.
\end{aligned} \tag{19}$$

Now, assume that $\mathbf{y} = (0, 0, 0)$. Since there is no information about such cells in the observed data, we have

$$N_{\mathbf{y}, g, x}^{(t)} = \mathbb{E}[N_{\mathbf{y}, g, x}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}] = m_{\mathbf{y}, g, x}(\boldsymbol{\theta}^{(t)}). \tag{20}$$

$\square$

5.2 M-step

Let us define

$$Q^{(Y)}(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}) := \mathbb{E}[\ell_c^{(Y)}(\boldsymbol{\theta}) \mid \{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}] \tag{21}$$

and

$$Q^{(V)}(\boldsymbol{\phi} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}) := \mathbb{E}[\ell^{(V)}(\boldsymbol{\phi}) \mid \{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}]. \tag{22}$$

We want to find

$$\begin{aligned}
(\boldsymbol{\theta}^{(t+1)}, \boldsymbol{\phi}^{(t+1)}) &= \arg \max_{(\boldsymbol{\theta}, \boldsymbol{\phi})} Q(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}) \\
&= \left(\arg \max_{\boldsymbol{\theta}} Q^{(Y)}(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}), \arg \max_{\boldsymbol{\phi}} Q^{(V)}(\boldsymbol{\phi} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\Pi}) \right).
\end{aligned} \tag{23}$$

Lemma 3. *The first maximization task is equivalent to fitting a Poisson Generalized Linear Model to the contingency table containing the values $\{N_{\mathbf{y},g,x}^{(t)}\}$.*

Proof. We have

$$\boldsymbol{\theta}^{(t+1)} = \arg \max_{\boldsymbol{\theta}} \sum_{\mathbf{y},g,x} [N_{\mathbf{y},g,x}^{(t)} \log m_{\mathbf{y},g,x}(\boldsymbol{\theta}) - m_{\mathbf{y},g,x}(\boldsymbol{\theta})]. \quad (24)$$

Under Assumption (A4), this formulation corresponds exactly to a GLM with a Poisson distribution and a logarithmic link function. $\square$

Lemma 4. *For the second maximization task, we maximize the expected conditional log-likelihood of the zero-truncated negative binomial distribution, i.e.,*

$$\boldsymbol{\phi}^{(t+1)} = \arg \max_{\boldsymbol{\phi}} \sum_{i \in S} \sum_{g=1}^9 \sum_{x=1}^2 \tau_{i,g,x}^{(t)} \log L_V(i | g; \boldsymbol{\phi}). \quad (25)$$

Proof. We have

$$\begin{aligned} Q^{(V)}(\boldsymbol{\phi} | \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}) &= \sum_{i \in S} \mathbb{E}[\log L_V(i | G_i; \boldsymbol{\phi}) | \mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}] \\ &= \sum_{i \in S} \sum_{g=1}^9 \sum_{x=1}^2 \left[\mathbb{P}(G_i = g, X_i = x | \mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}) \cdot \log L_V(i | g; \boldsymbol{\phi}) \right] \\ &= \sum_{i \in S} \sum_{g=1}^9 \sum_{x=1}^2 \tau_{i,g,x}^{(t)} \log L_V(i | g; \boldsymbol{\phi}). \end{aligned} \quad (26)$$

$\square$

5.3 Full Algorithm

1. **Input:** $\{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}, \hat{\boldsymbol{\Pi}}$.
2. **Initialization:** Initialize the weights $\tau_{i,g,x}^{(0)}$ and calculate the initial expected cell frequencies $N_{\mathbf{y},g,x}^{(0)}$.
3. For $t = 0, 1, \dots$, repeat until convergence:
 - (a) **M-step:** Obtain $\boldsymbol{\theta}^{(t+1)}$ and $\boldsymbol{\phi}^{(t+1)}$ by solving the maximization tasks described in Lemmas 3 and 4.
 - (b) **E-step:** Calculate the new weights $\tau_{i,g,x}^{(t+1)}$ and update the expected cell frequencies $N_{\mathbf{y},g,x}^{(t+1)}$ according to Lemmas 1 and 2.
4. **Output:** The final estimated parameters $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$, along with the weights $\hat{\tau}_{i,g,x}$ and the expected cell frequencies $\hat{N}_{\mathbf{y},g,x}$.

6 Total Number of Vacancies Estimator

For $i \in S$, we define

$$V'_{i,g} = \begin{cases} \tilde{V}_i, & \text{if } \tilde{V}_i = V_i, \\ \mathbb{E}_{\hat{\phi}}[V \mid V \geq 2, G = g], & \text{if } \tilde{V}_i = 2+, \\ \mathbb{E}_{\hat{\phi}}[V \mid G = g], & \text{if } \tilde{V}_i \text{ is missing.} \end{cases} \quad (27)$$

We estimate the total number of vacancies as follows:

$$\hat{T} = \sum_{i \in S} \sum_{g=1}^9 \sum_{x=1}^2 \hat{\tau}_{i,g,x} V'_{i,g} + \sum_{g=1}^9 \sum_{x=1}^2 \hat{N}_{(0,0,0),g,x} \mathbb{E}_{\hat{\phi}}[V \mid G = g]. \quad (28)$$

References

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- Brännäs, K. (1992). Limited dependent Poisson regression. *Journal of the Royal Statistical Society. Series D (The Statistician)*, 41(4):413–423.